

The Fine Print in the State Jobs Numbers

What three state-level labor-market datasets actually look like — before you model them

A data-validation walkthrough | continued claims • U3 unemployment • nonfarm payrolls | 50 states + DC, sourced from FRED / ALFRED

Three times a month — or, in one case, once a week — a fresh read on the American job market shows up on FRED, free for the taking. **Continued jobless claims** tick in weekly: a near-real-time count of people still drawing unemployment benefits. The **U3 unemployment rate** and **nonfarm payrolls** arrive monthly, the household- and payroll-survey twins that headline the national jobs report. Go to the state level and you get fifty-one of each — one per state plus D.C. — stretching back decades.

They are among the most-used inputs in applied macro: forecasts, lead-lag studies, backtests. And they look almost interchangeable — three tidy time series of the same labor market. They are not interchangeable, and the differences are exactly the kind that quietly break a model if you don't look first.

So before treating any of them as a feature, we put all three through the same four questions — call them V1 through V4 — using ALFRED, the point-in-time archive that remembers not just today's value but the value printed on the day it was first released. The questions are simple. The answers are where the three series part ways.

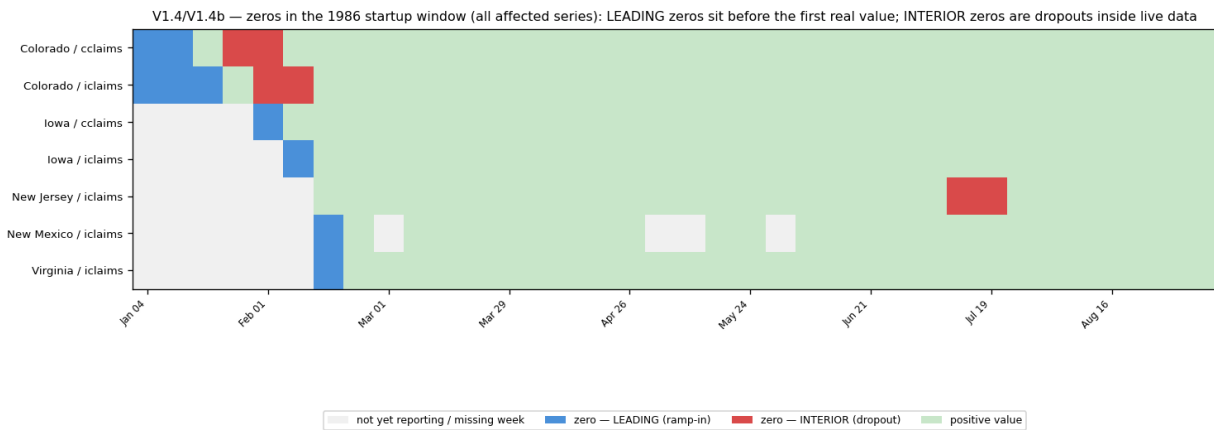
The series	Cadence	What it is
■ Continued claims	weekly	a count of people still claiming unemployment insurance
■ U3 unemployment	monthly	the headline jobless rate — a bounded percentage
■ Nonfarm payrolls	monthly	total jobs on employer payrolls — an unbounded count

We'll take the four questions one at a time.

V1 · Is the data even well-formed?

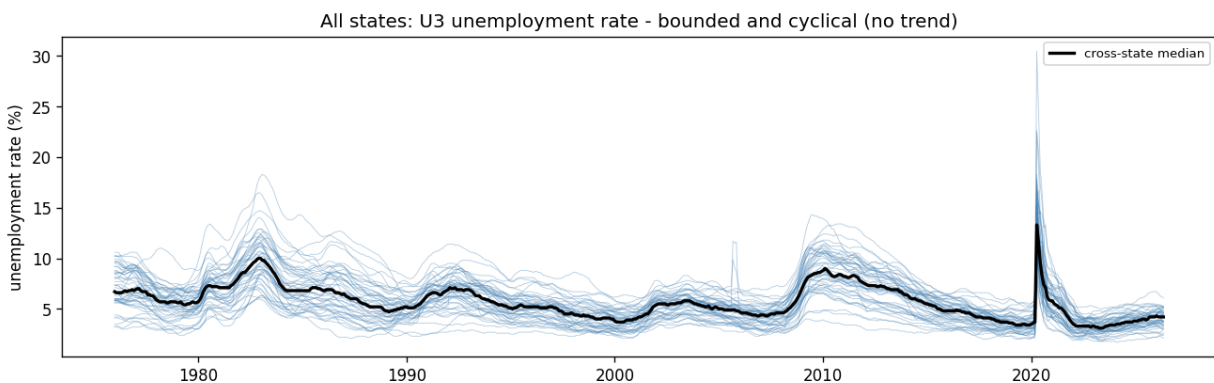
Start with the least glamorous question: can you trust the *shape* of the data before you trust its content? All three pass the basics — fifty-one states present, one value per period, no duplicate keys, no missing fields. The interesting part is what a 'valid value' even means for each.

Continued claims are counts, and counts should be positive. They almost always are — but the very start of the sample, back in 1986, hides a handful of zeros where a state hadn't begun reporting yet. Some sit at the front of a series (harmless — the series simply hasn't started); a couple sit in the *middle* of live data, which is a genuine dropout you must not quietly average through.



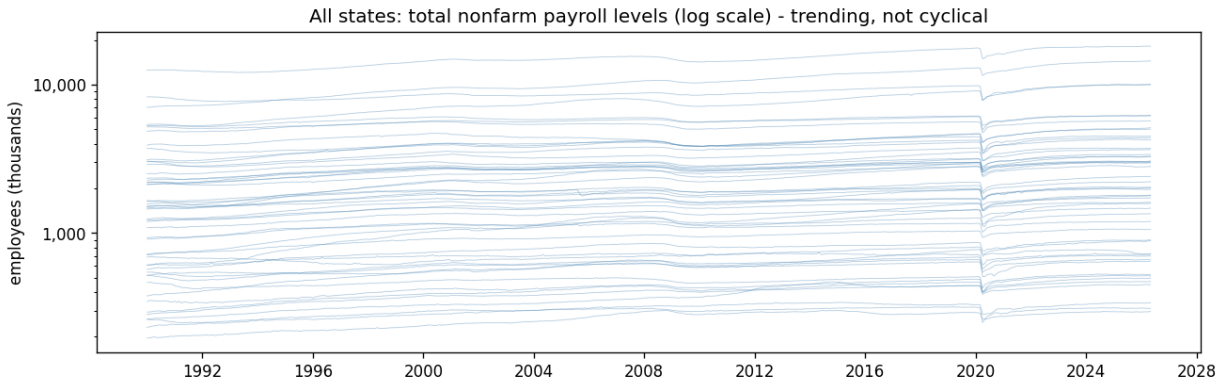
Claims, V1: the only zeros are 1986 start-up artifacts. Blue = leading (before a series begins, harmless); red = interior dropouts inside live data, which must be treated as missing.

The other two are different animals entirely, and it's worth seeing them side by side, because the contrast drives everything later. The unemployment *rate* is bounded — it lives between roughly two and thirty percent, quantized to a tenth of a point, and it *cycles*: up in recessions, down in expansions, never trending anywhere in particular.



U3, V1: a bounded, cyclical rate. Every state rises together in recessions (note the COVID spike to ~30%) and settles back — no long-run trend.

Nonfarm payrolls are the opposite: an unbounded count that mostly just grows. On a log scale, fifty-one states climb in near-parallel, punctuated by the 2008 and 2020 cliffs. One series mean-reverts; the other trends. That single distinction — bounded-and-cyclical versus unbounded-and-trending — decides how you'll transform each series before it ever reaches a model.



Payrolls, V1: an unbounded, trending count. The same labor market, a completely different mathematical object from the rate above.

V2 · What clock does it run on?

Every series runs on a calendar, and mismatched calendars are a classic source of silent bugs. Claims are weekly, each observation stamped to the Saturday that ends its week. The rate and payrolls are monthly, both referring to the pay period that contains the twelfth of the month.

Mostly the calendars are clean — no missing weeks or months once each series gets going. The exceptions are instructive. Claims have a scatter of gaps in the 1986–88 start-up years and one lone recent hole (Virginia, mid-2025).

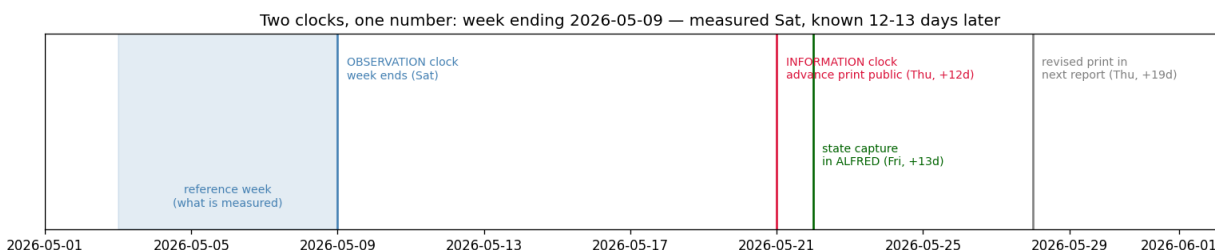


Claims, V2: missing weeks cluster in the noisy 1986–88 start-up era; the panel is otherwise complete right up to the present.

The most striking gap, though, is one all three share: **October 2025 is simply absent** — for every state, and for the national numbers too. That is not a fetch error. It is the footprint of the late-2025 federal shutdown, which delayed the releases. A good validation suite catches exactly this kind of thing and flags it rather than papering over it: don't interpolate across a hole that reflects a real break in data collection.

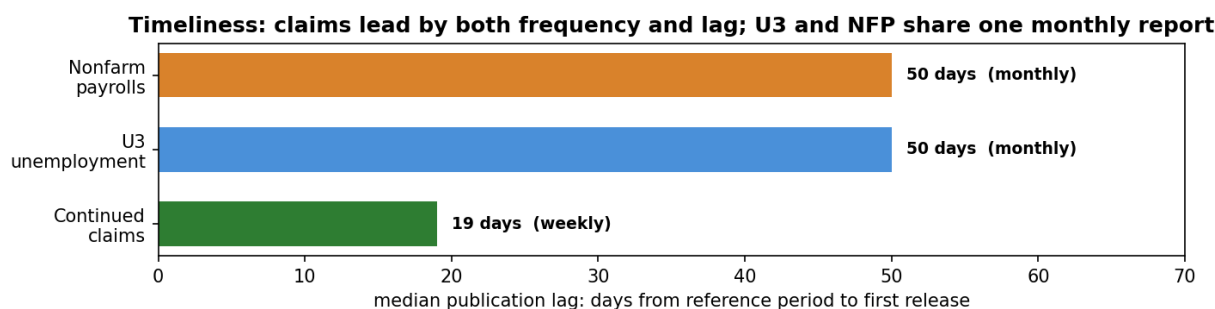
V3 · When do you actually learn the number?

Here is the idea that trips up more backtests than any other: **a number has two birthdays**. There is the period it measures — the week or month on the x-axis — and there is the day it actually became public, which can be weeks later. Confuse the two and you build a model that ‘knew’ things before anyone possibly could have.



V3, the two birthdays: a single week of claims is measured on its Saturday but only becomes public roughly two weeks later. Features must be joined on the second date, never the first.

The gap between the two birthdays is the **publication lag**, and it is where the weekly and monthly series split. Continued claims land about two weeks after the week they describe. The unemployment rate and payrolls take about seven weeks — and they arrive *together*, in the very same monthly *State Employment and Unemployment* report, roughly three weeks after the national jobs report you read about in the news.

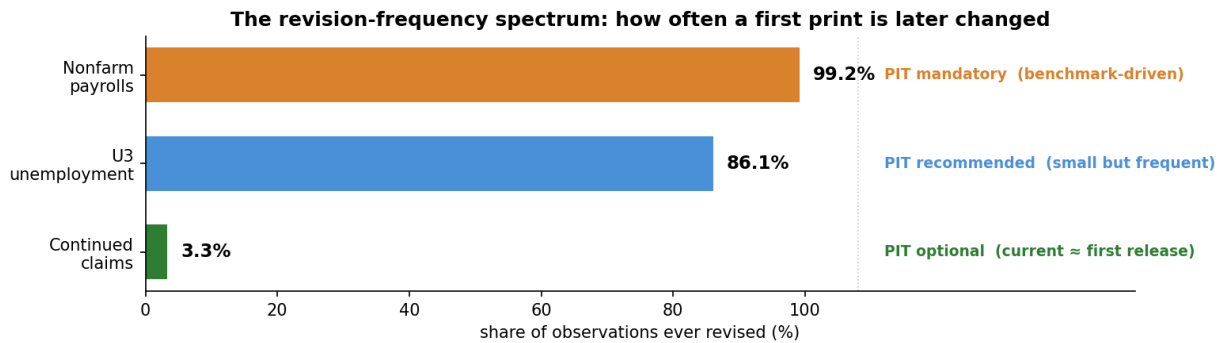


V3, timeliness: claims lead on both frequency and lag; the unemployment rate and payrolls share one slower monthly release.

That shared release is a small gift: because U3 and NFP for a given state come out on the same day, referencing the same reference week, they line up with each other for free. Claims, on their separate weekly cadence, need more care to align. And none of this point-in-time reconstruction runs forever — the archive that records first-print values begins around 2007–2010, which sets the earliest date any honest backtest can start.

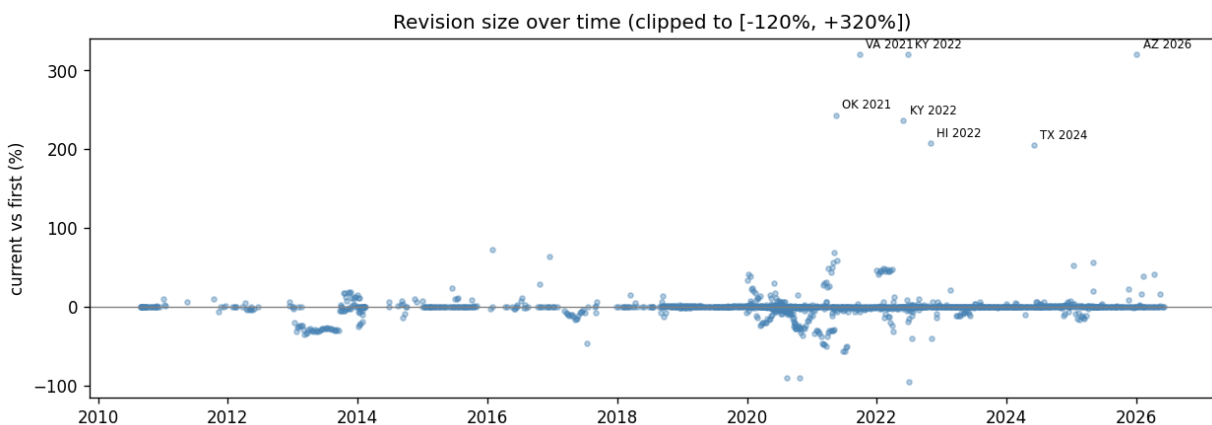
V4 · How much does the number change after you learn it?

Now the question that matters most, and the one where the three series stop resembling each other at all. When a number is first published, is that *the* number — or just a first guess that gets quietly rewritten later?



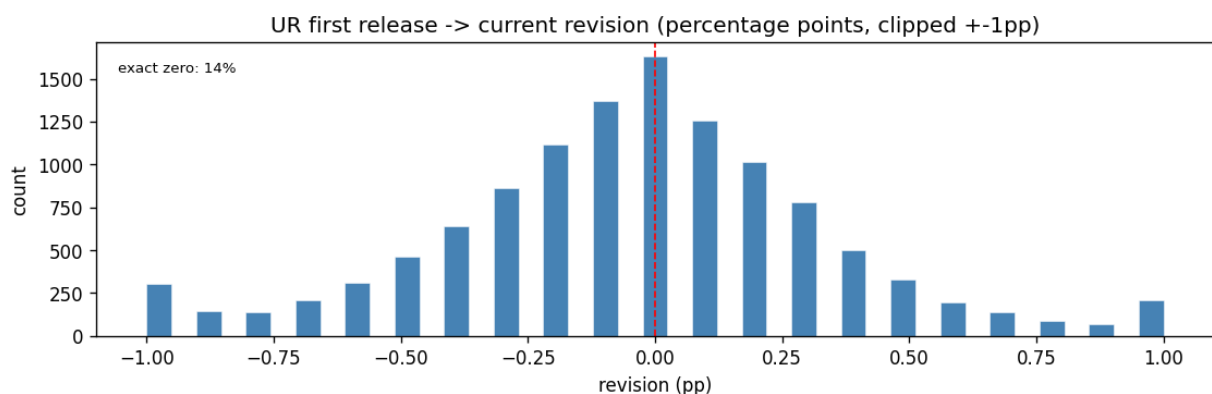
V4, the whole story in one chart: the share of observations ever revised after first release. Claims almost never change; payrolls almost always do. This single axis decides how hard you must work to avoid look-ahead.

The spread is enormous. **Continued claims are almost never revised**: about three percent of weekly values ever change, and when they do it is usually a one-off data-entry fix, not a systematic correction. For claims, the number you saw in real time is essentially the number you see today.



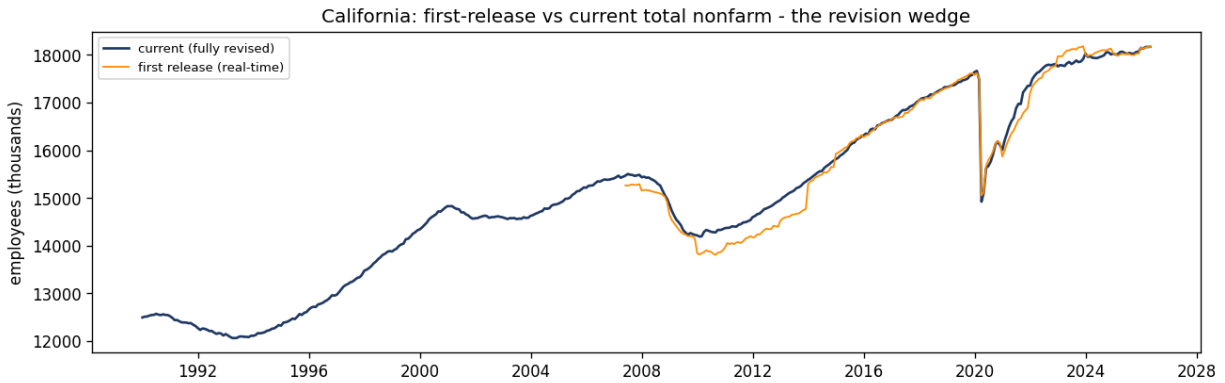
Claims, V4: revisions hug zero across fifteen years — the few outliers are labelled one-off input errors, not a pattern.

The unemployment rate sits in the middle. Most months do get revised — but the rate is published to only a tenth of a point, so the overwhelming majority of those revisions are tiny, rounding-level nudges. Benchmark years and the COVID shock produce the occasional move north of a full point, but day to day the first print is close to final.



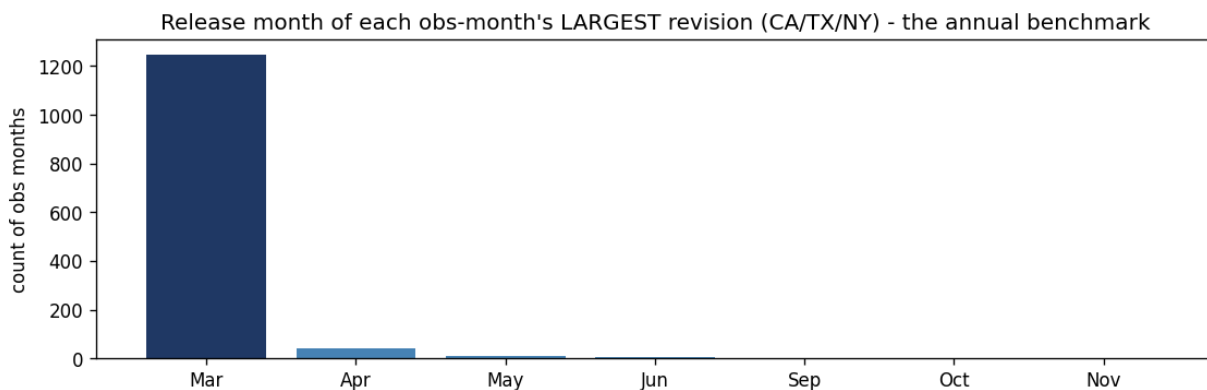
U3, V4: revisions pile up in the smallest bins — a tenth of a point either way — with only thin tails beyond half a point.

Nonfarm payrolls are the cautionary tale. Nearly every month is revised, and not by rounding. Each winter the whole recent history is re-anchored to a near-universal count of jobs — the annual benchmark — and the correction can run to hundreds of thousands of jobs in a big state. Watch California's real-time path pull away from its final, revised history:



Payrolls, V4: the revision wedge. The orange line is what you would have seen in real time; the navy line is the history as it reads today. The gap is what a non-point-in-time backtest silently leaks.

And those revisions do not trickle in randomly through the year — they arrive in a concentrated burst. Line up when each month's *largest* revision actually landed, and one bar towers over the rest: March, benchmark season.



Payrolls, V4: the release month of each observation's biggest revision. The March tower is the annual benchmark re-anchoring — apply it on its release date, never spread backward.

So what does this actually mean for a model?

Four plain questions, and the three datasets that looked interchangeable at the top of this post now sort themselves cleanly.

Transform each one for what it is. Claims and payrolls are counts — model them in logs or growth rates. The unemployment rate is a bounded rate — model it in percentage-point changes. Payrolls trend, so work with growth, not the level; claims and the rate cycle, so the level is fair game.

Respect the two birthdays. Index every feature by when it was knowable, not by the period it measures — and enforce that in proportion to how much each series is revised. For claims,

point-in-time data is a nicety. For the unemployment rate, it is advisable. For payrolls, it is non-negotiable: train on today's benchmarked history and your backtest is quietly cheating.

Aggregate with the right operator. To rebuild a national total, sum the counts — but you cannot sum rates. The national unemployment rate is a labor-force-weighted average of the state rates; the naïve average comes out biased low.

None of this makes the data bad. All three series cleared validation. It makes the data *specific* — and knowing exactly how is the difference between a feature that helps and one that leaks. That is what the fine print is for.